ARSET Remote Sensing

Earth is the planet NASA studies more than any other. Like our own bodies, the Earth is made up of connected, but different systems that interact in complex ways. A change is one system, for example, an increase or decrease in rainfall over time, can result in changes in one or more other systems, in this case changes in rainfall can impact the health of plants and crops, or the amount of of available groundwater. NASA studies the entire Earth System to not only observe what is changing, but also understand how changes in one of these systems impact all of the others.

Why Study Earth from Space?

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  Global Observations: Satellites help us collect information around the world, including places which are hard to observe from the ground like vast oceans, treacherous glaciers, and dense forests.

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  Reliability: NASA has long been a trusted source in scientific exploration. Our scientists and partners are dedicated to providing accurate and reliable data.

- •

  Long-Term Records: Most NASA satellites operate for many years, and even decades. Making routine measurements with the same instruments over these long time periods allow us to accurately track changes in Earth’s systems.

- •

  Data for Everyone: Not only scientists use this data. Professionals like doctors, teachers, farmers, journalists, and many others use satellite data to gain insights about the Earth to make more informed decisions in their work.

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Cooperation: NASA collaborates with agencies and organizations around the world to collect information on Earth systems for the benefit of all.

How Are Earth Observations Made?

With calibrated remote sensing instruments, scientists can measure a wide range of physical features for nearly every surface of the Earth's land, water, and atmospheric systems based on how these features interact with electromagnetic radiation.

Remote Sensing is the process of collecting information about objects from a distance, without being in physical contact with them.

Our eyes are an example of a remote sensing system. Human eyes detect red, green, and blue visible light, and our brains process and interpret this information into the full range of colors in the visible light spectrum. Artificial remote sensors, such as cameras and satellite instruments, use a similar process.

Remote sensing instruments are designed to detect specific sources of energy. They interpret these observations to provide data about objects on the Earth's surface and atmosphere.

Scientists refer to measurable properties within the Earth's systems as geophysical parameters.

Most of the remote sensors that NASA uses to study the Earth detect **electromagnetic radiation**.

Electromagnetic radiation is often referred to as light, electromagnetic energy, or electromagnetic waves.

Electromagnetic Waves
Electromagnetic waves can be described by their **frequency**, **wavelength**, and **amplitude**. All three describe a different property of light, yet they are related to each other mathematically.

Wave Characteristics

Source: ARSET

Frequency is the number of cycles (marked by crests) that pass a point in a given amount of time, usually per one second.

Wavelength is the distance from one wave crest to the next, and is inversely proportional to frequency. So, as frequency increases, wavelength decreases or gets shorter.
The amount of **energy** in a wave increases as the frequency increases, so waves with longer wavelengths have lower energy than waves with shorter wavelengths.

Amplitude is the measure of the wave from the **rest point** (center line) to the **crest** (the top of the wave) or **trough** (bottom of the wave). **Intensity** is a measurement used in remote sensing which is proportional to the square of the amplitude.

Wave Behaviors

Electromagnetic waves have predictable behaviors when interacting with the surfaces of different materials. Some ways that waves are affected when encountering an object are **reflection**, **absorption**, **scatter**, **transmission**, and **emission** (and re-emission) and are described below.

The incoming electromagnetic energy that hits a surface is called incident radiation.

Reflected energy refers to the electromagnetic waves that bounce off the surface at an angle equal and opposite to the angle of incident radiation.

Absorbed energy occurs when some of the incident radiation wavelengths hit a surface and are taken in by the object.

Scatter occurs when the surface causes incident radiation to bounce off the surface in multiple directions in varying wavelengths. A specific form is called **backscatter**, which occurs when energy scatters back toward the source.

Emission of specific wavelengths occurs when electromagnetic radiation is given off by the object from its own energy.
Another form is **Re-emission**, which occurs when some incident solar radiation is absorbed by an object and then emitted back, often at lower energy (longer wavelength).

Transmission occurs when the object is transparent to any wavelength of the incident radiation and allows it to pass through.

Polarization is another property of electromagnetic waves: the measurement of the electromagnetic field's alignment. The plane and direction in which the waves vibrate is their polarization. If all the waves are vibrating in the same direction, they are said to be “polarized” in that direction. Electromagnetic waves can be polarized or depolarized when they interact with materials and surfaces with certain textures.

The electromagnetic spectrum ranges from very long-wavelength, low-frequency, low-energy radio waves to very short-wavelength, high-frequency, high-energy gamma rays.

The electromagnetic spectrum describes the full range of wave frequencies that make up electromagnetic radiation.

Solar Radiation

Source: ARSET

The primary source of energy received on Earth is electromagnetic radiation emitted from the Sun. The Sun's radiation is reflected and absorbed by the Earth's atmosphere and surface. Satellites carry instruments that measure the electromagnetic radiation reflected or emitted.

The Sun emits most of its energy at optical wavelengths, between 0.1 μm to 4 μm. Solar energy, frequently referred to as **shortwave radiation**, includes ultraviolet, visible, and infrared radiation. In Earth satellite remote sensing, we are mostly concerned with wavelengths from ultraviolet (0.1 μm) through the microwave (100,000 μm).

Different wavelengths of the electromagnetic spectrum interact differently with the physical properties of materials that make up the Earth's surface and atmosphere.

Certain materials and physical properties have an individual and characteristic manner of interacting with incident radiation.

The **spectral response** (how the material reflects, absorbs, or emits radiation) is called the object's **spectral signature**. For example, certain materials will be more reflective (or their percent reflectance will be higher) than others at a given wavelength. The **percent reflectance** is the ratio of reflected energy compared to the total incident energy on the object at a given wavelength.

Radiance: A measure of the power of the electromagnetic radiation emitted from an object per unit area for a given wavelength. In the visible spectrum, it is also described as brightness of the source in a given direction. Radiance is directly proportional to the amplitude (and intensity) of the wave.

Surface Albedo: Is equivalent to percent reflectance, the fraction of incident radiation that is reflected by the Earth's surface for a given location and time. Values range from 0 to 1, where 0 means the surface absorbs all radiation and 1 means the surface reflects all radiation.

Vegetation

- The highest reflectance occurs in the near infrared (NIR) region. This range also has low absorption.
- In the visible range, reflection peaks at green wavelengths. Most incoming radiation in the visible range is absorbed through photosynthesis.

Water

- Longer visible wavelengths (green and red) are absorbed more by water than shorter visible wavelengths (blue) – so water usually looks blue or blue-green.
- There is low reflectance and high absorption in the infrared region.

Soil

- Soil is not highly reflective in any range. Reflectance increases gradually as wavelength increases.
- Absorption decreases as wavelength increases.
- Reflection and absorption are highly dependent on soil composition.

Ice/Snow

- Ice and snow are highly reflective at all visible wavelengths.
- There is lower reflection in the near infrared (NIR) region and very low reflection in short wave infrared (SWIR).

Atmosphere: Clouds

- Clouds are highly reflective for visible light.
- They are highly absorbent in the infrared region.

Atmosphere: Trace gases

- Much of the incident energy is absorbed and scattered by gases and aerosols in the atmosphere before reaching the Earth's surface.
- Most trace gases and aerosols relevant to air quality absorb in ultraviolet and visible ranges.

When designing a remote sensing instrument, engineers optimize the characteristics of the sensor to detect the spectral signatures of their desired targets, as well as distinguish between other spectral signatures which can impact the measurements.

All remote sensing instruments must strategically account for atmospheric interference on incident and outgoing radiation, including those observing terrestrial surfaces like vegetation and soil.

An Earth observing **satellite** is a spacecraft (also called a platform) which orbits around the Earth.

Remote sensing **sensors**, or **instruments**, that collect data about the Earth are mounted onto the satellite. The same type of instrument might be mounted on multiple satellites.

Sometimes, the terms “satellite” and “sensor” are used interchangeably, especially when a satellite has one main sensor that is critical to the science mission project, but they are not the same.

Land Satellite Science (Landsat 9) Mission
(opens in a new tab)

Date Launched: September 27, 2021 (9th Landsat mission in the series since 1972)

Instruments:
- Operational Land Imager 2 (OLI-2) for reflective band data
- Thermal Infrared Sensor 2 (TIRS-2) for the thermal infrared bands

Visible Infrared Imaging Radiometer Suite (VIIRS) Mission
(opens in a new tab)

Date Launched: October 28, 2011

VIIRS collects visible and infrared (IR) imagery along with global observations of Earth's land, atmosphere, cryosphere, and ocean

Instrument:
- Visible Imaging Infrared Radiometer Suite (VIIRS)

VIIRS is an **imaging radiometer** which includes detection of 9 bands in the visible and near-IR, 8 bands in the mid IR, and 4 bands in the long wave infrared (LWIR).

Tropospheric Emissions Monitoring of Pollution (TEMPO) Mission
(opens in a new tab)

Date Launched: April 7, 2023

Instruments: Tropospheric Emissions: Monitoring of Pollution (TEMPO)

TEMPO is a **spectrometer** that measures the intensity of hundreds of individual wavelengths in the visible and ultraviolet portion of the electromagnetic spectrum.

NASA-ISRO Synthetic Aperture Radar (NISAR) Mission

Launch Date: 2025

NASA-ISRO SAR Mission or NISAR is joint mission between NASA and the Indian Space Research Organization (ISRO) which uses a type of radar technology referred to as Synthetic Aperture Radar, or SAR.

Instruments: (polarimetric radars)

- L-Band Synthetic Aperture Radar (SAR)
- S-Band Synthetic Aperture Radar (SAR)

Satellites are categorized by their orbits and science applications, while sensors are identified by their energy source, resolutions, spectral ranges, and target applications.

Mission	**Landsat 9** (opens in a new tab)	**VIIRS** (opens in a new tab) **Land & Atm.** (opens in a new tab)	**TEMPO** (opens in a new tab)	**NISAR** (opens in a new tab)	**GPM** (opens in a new tab)
Satellite	Landsat 9	Suomi NPP	Intelsat 40e	NISAR	GPM
Example Sensor	OLI-2: Operational Land Imager 2	VIIRS: Visible Infrared Imaging Radiometer Suite	TEMPO: Tropospheric Emissions: Monitoring of Pollution	SAR: Synthetic Aperture Radar	GMI: GPM Microwave Imager

Two types of satellite orbits are most common for remote sensing of the Earth System: Geostationary Earth Orbit, or GEO, and Low-Earth Orbit, or LEO.

Geostationary Earth Orbit (GEO)
Satellites in Geostationary Earth Orbit (GEO) remain in a fixed position above the Earth. The satellite orbits in space around the Earth, above the equator, at the same rate as the Earth rotates around its axis.
In order for the satellite to orbit the Earth in this way, it has to be above the Earth's equator and roughly 36,000 km away from the Earth's surface. The GEO satellites are far enough away to view an entire hemisphere of the Earth it's facing, referred to as a "full disk view."

GEO Key Takeaways

1. 1Stays at a fixed position above the Earth's equator, rotating at the same rate
2. 2Distance from Earth is ~36,000 km
3. 3A full side of the Earth, the "full disk view", is always visible. Some sensors may view a smaller section referred to as its field of regard

Low-Earth Orbit (LEO)
Satellites in Low-Earth Orbit (LEO) are at altitudes between about 200 to 1,000 km, much lower than GEO satellites. Unlike GEO, which all occupy the same orbital path, there are several subcategories of LEO.

LEO Sun-Synchronous Key Takeaways

1. 1Satellite overpass always occurs at the same local time
2. 2Generally, one overpass for day and one for night (more overlap at high latitudes)
3. 3Polar-orbit allows for global coverage over a period of time

Some LEO polar-orbiting sensors have a narrow field of view due to their high spatial resolution. Hence, a given location on the Earth may not be observed twice a day, even though they are considered Sun-synchronous

LEO Polar-Orbiting and Non-Sun-Synchronous
Polar-orbiting satellites that are **non-Sun-synchronous** may just be referred to as "polar-orbiting." These orbits are chosen less often because Sun-synchronous orbits are advantageous for Earth observation, offering consistent lighting conditions. The main difference is that the polar-orbiting satellite does not maintain a consistent local solar time over its ground track as a Sun-synchronous satellite does.
LEO: Non-Polar (Inclined Orbit)
Unlike polar orbits, some LEO satellites have inclined orbits that result in irregular overpass times. This path allows for multiple overpasses each day for low- and mid-latitude regions.

The animation first shows the GPM satellite in a non-polar orbit, oscillating between 65 north and south latitudes. This is followed by the rest of the satellites within the GPM constellation, which includes both polar and non-polar satellite missions. Working together, this combination of polar and non-polar satellites results in a global precipitation estimate referred to as the Integrated Multi-satellitE Retrievals for GPM, or IMERG product, seen at the end of the animation.

GEO – Strengths & Limitations

Strengths
- Provides observations every few minutes, which is useful to capture rapidly changing events such as thunderstorms
- Valuable for constant operations of environmental (e.g., weather) and communications applications

Limitations
- Each GEO satellite sees only a portion of the Earth
- Lower quality observations of polar regions

LEO – Strengths & Limitations

Strengths

- Provide global coverage
- Observations for a given location are at the same local time
- High quality observations of polar regions

Limitations

- Less frequent observations than GEO, typically capturing a location one time per day and night, or less frequently (see Spatial Resolution and Coverage, Temporal Resolution subsections)

Mission	Landsat 9 (opens in a new tab)	VIIRS (opens in a new tab) Land & Atm. (opens in a new tab)	TEMPO (opens in a new tab)	NISAR (opens in a new tab)	GPM (opens in a new tab)
Satellite	Landsat 9	Suomi NPP	Intelsat 40e	NISAR	GPM
Example Sensor	OLI-2: Operational Land Imager 2	VIIRS: Visible Infrared Imaging Radiometer Suite	TEMPO: Tropospheric Emissions: Monitoring of Pollution	SAR: Synthetic Aperture Radar	GMI: GPM Microwave Imager
Orbit	LEO Sun-Synchronous	LEO Sun-Synchronous	GEO	LEO Sun-Synchronous	LEO Non-Polar

The sensor's energy source is a characteristic that is categorized as either passive or active.

Passive Sensors rely on external energy sources. They receive both reflected and emitted solar radiation during the day, and at night, they typically detect only emitted radiation.

Active Sensors provide their own energy source. The satellite carries a power source to emit radiation toward the Earth, and then a sensor is used to detect how much of that emission returns. Since active sensors produce their own radiation, they can be used day or night.

Passive sensors with detections in the UV, visible, and IR regions of the electromagnetic spectrum need a cloud-free view because clouds absorb or reflect at these wavelengths. Passive sensors with detections in the microwave regions can “see through clouds” since the longer wavelengths pass through the small cloud particles.

Active sensors can provide a spatial view of the scene, similar to passive sensors, but also the vertical distribution of retrieved geophysical parameters. View the animation to see the value of the GPM active sensor for the same example of Hurricane Francine shown above.

One method of active remote sensing is Light Detection and Ranging (lidar or LiDAR) which uses light in the form of a pulsed laser to measure the distances to the Earth from the satellite. This is useful for measuring the vertical distribution of features such as atmospheric pollutants, forest canopy structure, sea surface, and ice sheet heights.

Mission	Landsat 9 (opens in a new tab)	VIIRS (opens in a new tab) Land & Atm. (opens in a new tab)	TEMPO (opens in a new tab)	NISAR (opens in a new tab)	GPM (opens in a new tab)
Satellite	Landsat 9	Suomi NPP	Intelsat 40e	NISAR	GPM
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Orbit	LEO Sun-Synchronous	LEO Sun-Synchronous	GEO	LEO Sun-Synchronous	LEO Non-Polar
Energy Source	Passive	Passive	Passive	Active	Passive

The **spectral range** of an instrument represents the breadth of electromagnetic wavelengths detected by the sensor. Satellite instruments are designed to capture and quantify specific regions of the electromagnetic spectrum in discrete intervals. These intervals are called **channels** or **spectral bands**.

A sensor's spectral range is designed to address a particular set of science questions. The range of wavelengths included are not necessarily contiguous. They are included to detect the presence and/or absence of specific wavelengths. The design considerations to determine which spectral bands, or ranges of wavelengths, to detect include identifying the target as well as useful surrounding "secondary targets".

Desired Target's Spectral Signature - The range of wavelengths within the spectral signature of the target surface is the first consideration.

Secondary Object Spectral Signature - The range of wavelengths of secondary targets with similar spectral signatures that could interfere with the accuracy of the detection of the primary target surface.

Differentiator Spectral Signatures - For target surfaces with a wide spectral signature, a more narrow band is used to identify and differentiate secondary targets.

Typically, **multispectral sensors** observe at three or more bands, while **hyperspectral sensors** detect hundreds or thousands of very narrow bands (i.e., very high spectral resolution) to remotely sense very specific materials, such as gases in the atmosphere.

Mission	Landsat 9 (opens in a new tab)	VIIRS (opens in a new tab) Land & Atm. (opens in a new tab)	TEMPO (opens in a new tab)	NISAR (opens in a new tab)	GPM (opens in a new tab)
Satellite	Landsat 9	Suomi NPP	Intelsat 40e	NISAR	GPM
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Orbit	LEO Sun-Synchronous	LEO Sun-Synchronous	GEO	LEO Sun-Synchronous	LEO Non-Polar
Energy Source	Passive	Passive	Passive	Active	Passive
Spectral Range	Vis. NIR., SWIR, Panchromatic	Vis., NIR, SWIR, LWIR	UV, Vis., SWIR	Microwave ~24 cm (~1.25 GHz)	Microwave 0.16 to 3 cm (183 – 10 GHz)

The spatial resolution of a sensor represents the surface area of the Earth that is contained within one pixel in the resulting image from the remotely sensed data.

An image is a two-dimensional array of individual measurements arranged into columns and rows. These elements are called pixels, and the area of the pixel on the Earth's surface represents the spatial resolution within the image. Each pixel has two characteristics or pieces of information, an intensity and a location. The intensity value is the average of all emissions or reflectance from within the area covered by the pixel. The spatial resolution is often the same for each band of the sensor's spectral range, but there are sensors where the spatial resolution can vary by band.

Spatial resolution is typically given as a fixed value, but the actual size of an individual pixel will vary across an image due to the sensor viewing angle. The finest spatial resolution typically occurs when the pixel area is directly below the sensor, at nadir.

When a sensor view of the Earth is at an angle instead of directly overhead, the area covered is larger so the spatial resolution is coarser. This works like a flashlight aimed on the ground. The light beam covers more area if it is aimed at an angle rather than a point straight down.

Spatial Coverage: GEO Sensors

For geostationary sensors spatial coverage is typically the “full disk” of the Earth. However, some GEO sensors have scan strategies that limit the instrument view to only a portion of the entire full disk, often referred to as the “field of regard.”

The left image below shows the full disk of the Earth observed by the GOES-East satellite imager. In the right image, only the green outline portion of North America is observed by the NASA TEMPO mission.

GEO: Edge of Full Disk

The GEO satellite sensor view includes areas near the edge of the full disk where the sensor is viewing the Earth at a very steep angle. This is due to the larger viewing angle away from the nadir point, and also the Earth's curvature. This causes the GEO sensor observation near the edge to have a coarser spatial resolution compared to the center of the full disk.

Several GEO satellites placed strategically around the Earth would be able to minimize viewing angle and curvature impacts to spatial resolution in much of the image, but these sensors would still have a steep viewing angle towards the polar regions.

Spatial Coverage: Swath Width

LEO sensors orbit the Earth and view only portions of the surface beneath them. The spatial coverage that they observe is referred to as the "swath width", which is the total distance across the sensor's path.

The Landsat 9 Operational Land Imager-2 (OLI-2) has a swath width of 185 km, while other sun-synchronous missions, like the VIIRS imager, are 2000–3000 km in width. An extreme minimum in spatial coverage exists for Light Detection and Ranging (LiDAR) at less than 100 m, which are intended for vertical measurement of the atmosphere vs horizontal spatial coverage.

Spatial Resolution vs. Coverage

1. 1Generally, the wider the LEO instrument swath, the coarser the spatial resolution, and vice versa
2. 2Pixels at the edge of the swath for LEO sensors and the edge of the full disk for GEO sensors will have reduced spatial resolution due to larger viewing angles compared to pixels directly below the satellite at nadir
3. 3GEO sensor spatial resolution at the sensor scan edge will be more impacted than LEO sensors due to the curvature of the Earth
4. 4Polar LEO is necessary for capturing the entire globe, including the poles

Mission	**Landsat 9** (opens in a new tab)	**VIIRS** (opens in a new tab) **Land & Atm.** (opens in a new tab)	**TEMPO** (opens in a new tab)	**NISAR** (opens in a new tab)	**GPM** (opens in a new tab)
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Orbit	LEO Sun-Synchronous	LEO Sun-Synchronous	GEO	LEO Sun-Synchronous	LEO Non-Polar
Energy Source	Passive	Passive	Passive	Active	Passive
Spectral Range	Vis. NIR., SWIR, Panchromatic	Vis., NIR, SWIR, LWIR	UV, Vis., SWIR	Microwave ~24 cm (~1.25 GHz)	Microwave 0.16 to 3 cm (183 – 10 GHz)
Spectral Resolution	9 Bands	22 Bands	700 Bands (Hyperspectral)	1 Band	13 Bands
Spatial Coverage	185 km Swath	3000 km Swath	N. America (Daytime)	240 km Swath	885 km Swath
Spatial Resolution	0.015 km, 0.030 km (15 m, 30 m)	0.375 – 0.750 km	2 x 4.5 km	0.003 – 0.010 km (3-10 m)	6 – 25 km

LEO Polar Orbit – Wide Swath; Better Temporal Resolution
For polar-orbiting LEO instruments with a wide swath, the temporal resolution for most locations is 1 day, in terms of daytime and nighttime observations. However, some low latitude locations could have gaps if the swath width does not overlap with each orbit.

LEO Polar Orbit – Narrow Swath; Limited Temporal Resolution
Other polar-orbiting LEO instruments with very high spatial resolution tend to have narrow swaths, and thus very large gaps in satellite observations occur between orbits. This decreases the temporal resolution or increases the revisit time to several days to weeks depending on the instrument's swath width.

LEO Non-Polar Orbit
In a LEO non-polar orbit, some high-latitude locations will not be observed. Again, the swath width and orbit path will determine the temporal resolution. In some cases, a series of LEO satellites with the same instrument are placed in an inclined orbit in order to observe tropical and mid-latitudes more often. While the temporal resolution for an individual satellite may be 1–2 days, the effective temporal resolution by the suite of LEO satellite instruments in a series becomes much less.

GEO Temporal Resolution
For GEO satellite instruments, the temporal resolution is generally much higher (less amount of time between observations) compared to LEO instruments. Additionally, the temporal resolution of GEO imagers can vary due to the ability to perform multiple scans within the same given time.

Mission	Landsat 9	VIIRS	TEMPO	NISAR	GPM

	(opens in a new tab)	(opens in a new tab) **Land & Atm.** (opens in a new tab)	(opens in a new tab)	(opens in a new tab)	(opens in a new tab)
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Spatial Resolution	0.015 km, 0.030 km (15 m, 30 m)	0.375 - 0.750 km	2 x 4.5 km	0.003 – 0.010 km (3-10 m)	6 – 25 km
Temporal Resolution	16-Day	0.5- to 1.0-Day	Hourly	6-Day	0.4- to 1.6-Day (Latitude Dependent)

Higher radiometric resolution allows for more detailed differentiation between various surface materials and conditions, increasing precision. This can be crucial for applications like land cover classification, vegetation analysis, and monitoring environmental changes.
As an example of a sensor's radiometric precision, consider two thermometers: one that measures temperature in only whole degrees Celsius from 1 to 100° C (i.e., a hundred values). Now imagine a more sensitive thermometer that measures temperatures to a tenth of a degree Celsius. The second thermometer has higher precision (note: not the same as accuracy). For this improved precision, the thermometer itself has to be able to differentiate between 1000 magnitude values. This same type of "sensitivity" for satellite instruments exists in the form of radiometric resolution.
This means that the sensor's radiometric resolution impacts the quality of the data collected and the types of analyses that can be performed.

he raw satellite instrument measurements are stored electronically in data files that are classified by their bit memory storage. A bit is simply a binary value of either 0 or 1. All data is stored as a group of bits (like a computer). The greater the number of bits used, the greater the radiometric resolution.

For example, a 2-bit data file only has 4 possible magnitude combinations (i.e., $2^2=4$), and the left image represents an image that only uses 4 shades of gray. This results in a more uniform appearance to the image and a lack of details (*click on the image to enlarge*).

However, a 4-bit image has 16 possible magnitude combinations (i.e., $2^4=16$), and the center image has more detail because a greater number of values are available to define each pixel of the image, which provides more contrast of features. Thus, land features have more detail due to increased variation in the imagery. This provides more continuity of details within the imagery.

Lastly, the 8-bit image on the right can hold 256 (i.e., $2^8=256$) possible magnitude value combinations per pixel , and a very smooth scale of black to white defines the image to provide high contrast of features. Very subtle features in the imagery become apparent, as seen in the lower right.

A higher radiometric resolution requires more bits to store the data, and results in sharper imagery details.

Mission Science Drives Decisions
We might assume that every satellite and sensor is designed with the highest functionality:

Large Spectral Range
I.e., covers as much of the EM Spectrum as possible

High Spectral Resolution
I.e., each band from the sensor is impacted by a very narrow set of wavelengths

Wide Spatial Extent
I.e., a large area is seen by each sensor scan/swath

High Spatial Resolution
I.e., each pixel by the sensor sees small-scale details

High Temporal Resolution
I.e., the sensor quickly provides new scans of the same area

High Radiometric Resolution
I.e., small differences in energy per band are retained in the data

Trade-Offs are Necessary
In general, as technology advances, satellite instruments are able to achieve better and better performance. However, technical and physical limitations require trade-offs between these characteristics. Whenever an instrument is being designed, these tradeoffs will be guided by available technology, cost, and the priorities of the mission planners.

PACE has a low-Earth orbit (LEO) sun-synchronous orbit which means it is a polar-orbiting satellite that will view the Earth's poles daily, and in many areas of the poles PACE passes over multiple times per day. Thus, PACE has an advantage over geostationary orbits which do not have a viewing angle to view the poles. The temporal resolution of PACE is stated as 2 days, but this is due to small gaps in coverage near the equatorial regions which require 2 days of orbiting to achieve full global coverage. Due to the LEO sun-synchronous orbit and wide swath width like VIIRS, the polar regions will be observed each day.

While both PACE and GMI are passive sensors, they operate in different parts of the electromagnetic spectrum. Additionally, the PACE spatial resolution is much better than GMI, and thus, PACE could be used to observe events that near a kilometer wide, but GMI could not be used for the same scale of events. Lastly, PACE is polar-orbiting while GMI is non-polar; therefore, GMI would not be able to observe the poles but PACE would provide data over the poles.

PACE provides global coverage every 1-2 days (depending on latitude) and thus could detect features such as fog, dust storms, and volcanic ash as well as monitor vegetation health using its visible and infrared channels. Since thunderstorms evolve on hourly time-scales, the daily temporal resolution of PACE is not sufficient to monitor their changes. While PACE resolution is good compared to geostationary sensors, changes in land use at the farmer's field scale would require spatial resolutions more like imagers such as OLI on Landsat-9 that can provide information at 15-30 meters.

The remotely sensed data from satellite sensors is typically visualized with satellite imagery. This imagery can represent one channel or a combination of several channels from the sensor. While the satellite sensor data itself is quantitative, the imagery is created by mapping this data to a range of colors for qualitative visual analysis.

Single Channel Imagery
For each single band, satellite remote sensors directly measure **radiance**, which is the intensity of radiation in a given direction. Radiance is often quantified using a Brightness Temperature scale, expressed in degrees Celsius. The GOES "Clean Infrared" channel (or Band 13) is shown here as an example, and a color bar is applied to the data in order to highlight the cloud top surfaces with the lowest thermal temperatures. The red to magenta color range corresponds to the coldest cloud tops and strongest convection. Note also the warm cloud, land, and water surfaces that are less prominent in the light gray shades, and a different color curve could be used instead to highlight these warm features in this same scene. This single band can be used for analysis on its own, or combined with other bands in more complex imagery.

Given measurements of radiance, the ratio of the radiation striking a surface to the radiation reflected off can then be calculated, and this is referred to as **reflectance**.

One way that scientists create more complex satellite imagery is to combine reflectances from multiple bands.

True Color Imagery
True Color imagery was designed to display the Earth in colors similar to what we might see with our own eyes. The product is primarily a combination of available channels that are sensitive to the red, green, and blue

visible light. The image below shows how the VIIRS instrument's visible red, green and blue bands are combined to generate the composite true color image (i.e., a Red-Green-Blue, or RGB imagery product)

False Color Imagery

While true color images are created by assigning the red, green, and blue colors to the red, green, and blue visible wavelengths, respectively, false color images are created by assigning different wavelengths to one or more colors. For example, we can use color to depict non-visible wavelengths, such as infrared or ultraviolet which may be associated with things like atmospheric gases and heat, to create an enhanced image for analysis. This is why the image colors may not correspond to what we expect to see.

A color bar defines a specific pixel color to be used for each of the data values within the imagery/product. For example, a satellite brightness temperature (Kelvin) difference of +5 K is defined to have a yellow pixel color vs -5 K is defined to have a blue pixel color.

Alternatively, when using an RGB imagery product, each pixel contains a red, green, and blue color component toward the final pixel color. A satellite band or band difference is used within each color component of the RGB in order to identify multiple geophysical parameters within one image (i.e., cloud thickness, phase, and height all within the same image).

Then, the geophysical parameter range is scaled between black (i.e., no color) to the maximum color intensity per RGB component (i.e., full red, full green, or full blue).

— Except for the True Color imagery product that mimics what you would see with your eyes if you were in space, all other RGB imagery products are False Color. In other words, the colors that represent features in the imagery are not the same colors your eyes would see in reality. RGB imagery is not the same as applying a color bar to satellite imagery, but rather it is the application of bands to specific color inputs of red, green, and blue to a resulting final imagery product.

Progression of Satellite Data Processing

From raw sensor detections to geophysical products, there are many steps that go into creating satellite data products. NASA describes these steps using Data Processing Levels. Users will want to choose data from these levels depending on their experience and intended application. As the data processing level increases, more analysis is being done. This can make the data easier to understand and work with, but can also obscure certain nuances of the data.

We start at Level-0, with the raw measured radiance from the satellite sensor itself. This data is unprocessed and at the full resolution of the satellite’s sensors. While most casual users of the data rarely interact with the Level-0 information, this is the starting point from which science teams work to create all other higher-level satellite data products

At Level-1, we use information on the position of the satellite and the the sensor view to reference the Level-0 raw data to the corresponding locations on Earth where the sensor was pointing. This is referred to as "Georeferenced data." This image shows the single-channel, brightness temperatures measured by the VIIRS sensor, geolocated on the Earth. Darker shades correspond to clouds or colder surfaces.

Where we start to get more useable data is at Level-2, when we apply **retrieval algorithms** (i.e., mathematical formulas) to the remote sensing data to derive geophysical quantities (e.g., temperature, precipitation, ocean color, wildfire intensity, etc.). In this example, we use an algorithm on Level-1 data to derive an index to measure the scattering and absorption of visible wavelengths by smoke, dust, and other particulates in the atmosphere (*see image*). A user applies this Level-2 derived index to analyze the potential impacts of pollutants in the atmospheric column. Note the white areas; these show **data gaps**, or areas where the retrieval algorithm was unsuccessful, for example due to the presence of clouds or very reflective surfaces. At Level-2, the data is still at the pixel resolution of the satellite, which is the highest resolution we can get. However, the locations and sizes of pixels will not necessarily be consistent, depending on the orbit of the satellite and the angle between the sensor and the Earth.

To make the data easier to work with, we often also create Level-3 datasets. These take the Level-2 data and **translate them to a uniform grid** in space and time; for example, a latitude-longitude grid over the Earth's surface, with all data collected over a full day being aggregated together to produce one map, as shown here. As part of this process, we also typically do some **quality control**, screening out pixels where there was too much cloud cover or where the retrieval algorithm performed poorly for some other reason, in order to get the best quality data product. There is a trade-off here: in making Level-3 data, we often need to average together several pixels from the Level-2 data, usually resulting in a coarser resolution but more accurate product.

We also sometimes create Level-4 data products, which combine the satellite data with other sources of information, especially models. This allows us to do things like fill in the gaps in the data with model estimates or make inferences about geophysical quantities which we cannot retrieve from the satellite data directly. Since we're now bringing in these other sources of information, which involves making assumptions, the Level-4 data can have greater uncertainties compared to satellite-only data, but it can still be useful in situations where complete data coverage in space and time is important.

- L0: Raw Radiance – Unfiltered detections by sensor
- L1: Geolocated and Calibrated – Brightness data per channel
- L2: Algorithm-Derived Fields from L1 – Geophysical parameters such as precipitation
- L3 Gridded – Spatially and/or temporally averaged and mapped onto a uniform grid
- L4: Satellite Combined with Other Sources – Often model outputs are integrated for derived

Typically we recommend new users to start with Level-3 data since it is easier to use because it's on a uniform grid. A downside, of gridded data is that it typically reduces the spatial resolution, and different gridding algorithms can produce slightly different estimates for the grid cell. So, for applications needing very high spatial resolution, it may be necessary to go back to the Level-2 data.

Accessing Remote Sensing Data and Products

All NASA data is available to the public at no cost through the NASA Earthdata (opens in a new tab) website.

NASA uses Distributed Active Archive Centers (DAACs) throughout the US to process, archive, and distribute NASA Earth observation data, and support data users with analysis tools and technical assistance.

Through the Earthdata portal, users can access and view satellite remote sensing imagery data and tools for visualizing and analyzing data.

The Shape of the Earth: Geoid

Although it is commonly thought of as a sphere, Earth is not perfectly spherical. Its actual shape is what we refer to as a “geoid.”
A geoid is a modeled shape of the Earth's gravity field representing mean sea level (if tides, winds, and other factors did not exist) and its imagined extension under (or over) land areas.

For spatial data to be displayed in a spatially consistent way, we use an elliptical spheroid to approximate the surface of the Earth.
No spheroid is a perfect fit, so many different approximations are used. Each approximation will fit one part of the Earth’s surface better than others. Each of these spheroids is calculated using a specific "datum" as a reference point.

A **datum** is a known point on Earth’s surface or within its geometry that we can use as a reference point for all other locations.
Because of the irregular shape of the planet, the use of datums is necessary to portray spatial data as accurately as possible.

All spatial data, including satellite imagery, must be indexed, or **georeferenced**, to a fixed point on Earth’s surface.

Pairing a data point or pixel with a specific location on the ground requires a **coordinate reference system (CRS)**.

There are two types of commonly used coordinate systems: **geographic** and **projected**.

Geographic Coordinate Systems -

Pros:
- Better for the entire Earth
- Good for data with a large spatial extent

Cons:
- Less accurate for specific regions
- Bad for data with a small spatial extent

Projected Coordinate Systems -

Pros:
- Better for specific locations
- Good for data with a small spatial extent

Cons:
- Accuracy is skewed to one specific location

- Not useful for data with a large spatial extent

Types of Projected Coordinate Systems

Cylindrical (Example: UTM)
- Best for use in Equatorial Regions
- Distortion occurs near poles
- Great for medium spatial extents

Conical (Example: Albers Equal Area Conic)
- Best for use in Polar regions
- Distortion increases farther from pole

Planar (Example: State Plane Coordinate System)
- Also known as “Azimuthal”
- Projects data onto a flat plane
- Great for small spatial extents

True: From interpreting raw level 0 and 1 data through sophisticated level 4 blended with non-satellite data sets, the different levels of satellite data meet the needs of users at all levels of experience and knowledge with remote sensing.

a few examples of how satellite data is being used in real world applications.

PRECIPITATION MONITORING
VEGETATION MAPPING
AIR QUALITY TRENDS
WILDFIRE DETECTION

Ten **LEO** satellites make up the Global Precipitation Measurement (GPM) Constellation.

Over time, the entire Earth is mapped by the numerous overpasses of the satellite sensors, and we get a global view of precipitation by **using the spectral signature of water and ice within the microwave portion of the electromagnetic spectrum**.

Satellite sensors with **high spatial and spectral resolutions in the near-IR and visible** portions of the electromagnetic spectrum can tell us information about vegetation and crop health.

Sun-synchronous LEO satellite orbits offer the opportunity to examine long term changes for a given location due to their observations at the same time each day. Thus, impacts from federal environmental laws like the Clean Air Act (1970) can be assessed via temporal composites from these satellite sensors.

Sensors with **hyperspectral resolution in the visible and UV** wavelengths are used to derive **Level-2** geophysical parameters such as the amount of NO2 in the atmospheric column. NO2 is a pollutant that is produced during combustion in gasoline powered engines and coal-burning power plants. NO2 can negatively impact the respiratory system, and it contributes to the formation of other pollutants like ozone.

To examine changes over long periods, the daily, instantaneous retrievals from **LEO** satellite overpasses can be composited into **monthly or yearly NO2 Level-3 products**. If we combine this Level 3 data with additional data (from atmospheric computer models, for example) to estimate surface-based NO2, we would have a **Level-4** dataset.

Satellite **bands in the near-IR (NIR) and shortwave-IR (SWIR) wavelengths** are used to study and monitor wildfires. Energy from wildfires is so intense compared to the surrounding land that the sensors can detect an abnormally large amount of radiance in Level-1 data.

Since the different sensor channels detect different parts of the **spectral signature**, the **near infrared (NIR) and shortwave infrared (SWIR)** are used together to derive the intensity of the fire (**Level-2** product), referred to as fire radiative power (FRP).

Navigating Worldview

Layers, Events and Data Panel

Manage the map imagery in this panel through the Layers, Events and Data tabs:

- **Layers Tab**:
    - **Layer List**: shows all the imagery layers that are loaded on the map. Base layers appear under map overlays (not shown in this image). Toggle on and off the layers by clicking on the "eye" icon to the left of the layer name. Only the topmost active base layer displays. To reorder layers in the list, click and hold on a layer or group and move it in the Layer List.
- **Events Tab**: Select the Events tab to view a curated list of current and historical natural events around the world to load onto your map.
- **Data Download Tab**: Select the Data tab to download the underlying data for the imagery layers displayed on the map. You will be directed to the Earthdata Search tool to download data granules of the selected layer, chosen area of interest, and current date.

Layer Picker

- **Layer Picker**: Click the Add Layer button to browse and search over 1000 imagery layers by science discipline, hazards and disasters categories, or by keyword.
- **Start Comparision:** Compare imagery from two different dates, or two sets of imagery from the same date or even two sets of imagery from different dates.

Timeline Management, from left to right:

- **Date Selector:** Change the date (and time of day if you have Geostationary layers loaded) by typing into the date fields, or by clicking on the up and down arrows by each field.
- **Interval Selector:** Select the desired timeline interval and use the left and right arrows to step backward or forward in time according to the selected interval. Use the far right arrow to go to the latest available date.
- **Animation:** Animate the imagery by clicking on the video camera icon. Click on the Play button to watch the imagery change. You can also select an area to export as an animated GIF.

- **Timescale Adjuster:** Adjust the timescale to years, months, and days (and, hours and minutes if you have a Geostationary layer loaded).
-

In the Worldview map below, the 'comparison mode' is used to show different layers for the same day, August 8, 2021. The base layer on Side 'A' (left) shows the VIIRS (on the NOAA-20 satellite) true color imagery. The base layer on side 'B' shows VIIRS (NOAA-20 satellite) false-color imagery.

To have many observations per day at the same location, an instrument on a geostationary satellite is needed, such as the GOES ABI, but global measurements require a polar-orbiting satellite and wide swath width such as the VIIRS instrument on the NOAA satellite series. For the height of a feature such as a smoke or ash plume, an active sensor is required such as the CALIOP lidar.

Aerosols interact with radiation in the optical parts of the electromagnetic spectrum. Their presence and optical properties can be measured with multispectral sensors.

When measured with in-situ sensors, aerosols are typically classified by size, and their presence is quantified using size-based measurements like **$PM_{2.5}$** and **PM_{10}**. Aerosols can also be classified by their chemical composition, where their presence is quantified by a type-specific mass concentration.